# Notes on lecture 13

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## 1 Triple integrals

Recall that in order to perform double integral of $f(x, y)$ over a region $D$, we have seen that it all boils down to defining the region $D$ accurately. Similar case is going to happen for triple inetgrals. Let us start with the simplest possible case, where the region in $\mathbb{R}^3$ is going to be such that the $x, y$, and $z$ coordinates lie within some intervals. That is, the region is a rectangular box.

Consider a region $B$ ($\subseteq \mathbb{R}^3$) of the form of a rectangular box. Say,

$$B = \{(x, y, z)|a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}.$$

Let $f(x, y, z)$ be a function of 3 variables and suppose that we need to compute the integral $\int\int_B \int f(x, y, z)dV$ (Here $V$ stands for "volume". We will mention later on about what dV represents.). The graph of $f$ can't be visualized because $f$ is a function of 3 variables.

As in the 2-dimensional case, let us divide the intervals $[a, b]$, $[c, d]$, and $[r, s]$ into $l, m$, and $n$ many subintervals respectively. Say, $[a, b]$ is divided as

$$a = x_0, x_1, \ldots, x_l = b.$$

Similarly, say $[c, d]$ is divided as

$$c = y_0, y_1, \ldots, y_m = d,$$

and $[r, s]$ is divided as

$$r = z_0, z_1, \ldots, z_n = s.$$

As before, these subintervals need not be equally spaced.

Then we are going to get a grid as shwon in Figure 1.0.1. From this grid, we

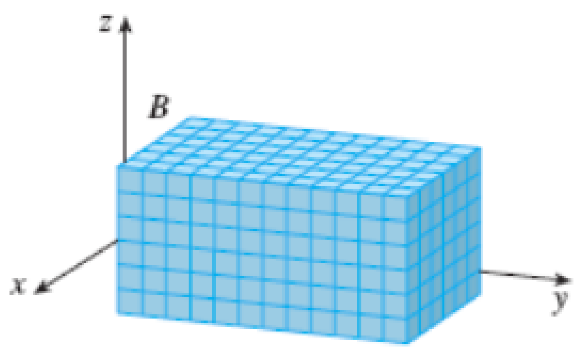


Figure 1.0.1: The rectangular box $B$ partitioned into a grid

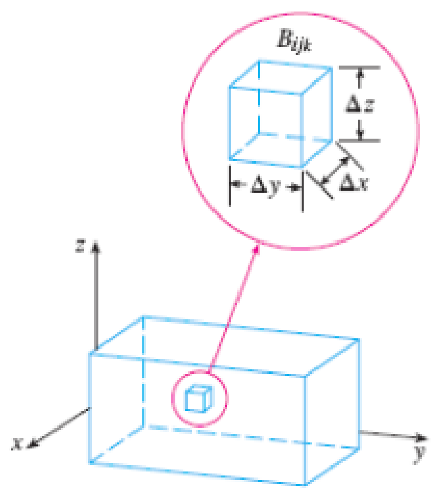


Figure 1.0.2: The small rectangular box $B_{ijk}$

are going to choose a small rectangular box $B_{ijk} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k]$. Say, this small rectangular box $B_{ijk}$ has dimensions given by $\Delta x, \Delta y$, and $\Delta z$. Please see Figure 1.0.2 for this. Let $\Delta V$ $(= \Delta x \Delta y \Delta z)$ be the volume of $B_{ijk}$. Let $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ be a sample point in this small rectangular box $B_{ijk}$. Then we form the **triple Riemann sum** as

$$\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)\Delta V.$$

The **triple integral** of $f$ over the entire rectangular box $B$ is then defined by

$$\int\int_B\int f(x,y,z)dV = \lim_{l,m,n\to\infty}\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)\Delta V$$

if the limit exists.
As we let $l,m,n \to \infty$, we can take any point in $B_{ijk}$ as the sample point $(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)$. In particular, we may take the sample point $(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)$ to be the corner point $(x_i, y_j, z_k)$. Then we have

$$\int\int_B\int f(x,y,z)dV = \lim_{l,m,n\to\infty}\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_i, y_j, z_k)\Delta V.$$

**Theorem 1.0.1.** ***Fubini's theorem for triple integrals:*** *If $f$ is continuous on the rectangular box $B = [a,b]\times[c,d]\times[r,s]$, then*

$$\int\int_B\int f(x,y,z)dV = \int_r^s\int_c^d\int_a^b f(x,y,z)dxdydz.$$

## 1.1 Triple integrals over general bounded regions

For double integrals, recall that we had 2 types of regions, namely, type I and type II. We also studied that any bounded region (in $\mathbb{R}^2$) can be expressed as a union of type I or type II or both type of regions.
Likewise here, we are going to have three types of regions, denoted by type 1, type 2, and type 3.

**Remark 1.1.1.** For triple integrals, we use Arabic numerals 1,2,3 to denote the three types of regions. Whereas, for double integrals, we had used the Roman numerals I and II.

A **type 1** region in $\mathbb{R}^3$ is of the form

$$E = \{(x,y,z)\in\mathbb{R}^3 | (x,y)\in D\subseteq\mathbb{R}^2, u_1(x,y)\le z\le u_2(x,y)\},$$

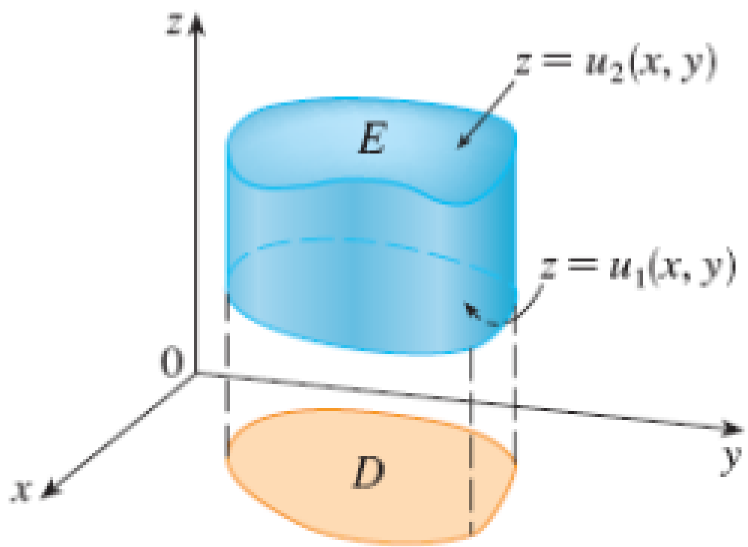


Figure 1.1.1: A type 1 solid region

which is bounded by two surfaces that are graphs of functions of the variables $x$ and $y$. Figure 1.1.1 gives an illustration of a type 1 region in $\mathbb{R}^3$.

If we integrate a continuous function $f(x, y, z)$ over the above mentioned type 1 region $E$, then we will have

$$\int\int_E\int f(x,y,z)dV = \int_D\int \left[\int_{z=u_1(x,y)}^{u_2(x,y)} f(x,y,z)dz\right] dA.$$

The expression $\int_{z=u_1(x,y)}^{u_2(x,y)} f(x,y,z)dz$ (inside the paranthesis) in the right hand side of the above equation will be a function (say, $\phi(x,y)$) of $x$ and $y$. Then we will have

$$\int\int_E\int f(x,y,z)dV = \int_D\int \phi(x,y)dA,$$

where the integration of the right hand side expression can be easily performed by using techniques of double integration from the earlier lectures. The plane region $D$ is going to be a union of type I or type II or both types of plane regions.

If $D$ is a type I plane region, then

$$E = \{(x,y,z)|a \leq x \leq b, g_1(x) \leq y \leq g_2(x), u_1(x,y) \leq z \leq u_2(x,y)\}.$$

Figure 1.1.2 gives an illustration of such a region.
In this case, the integral

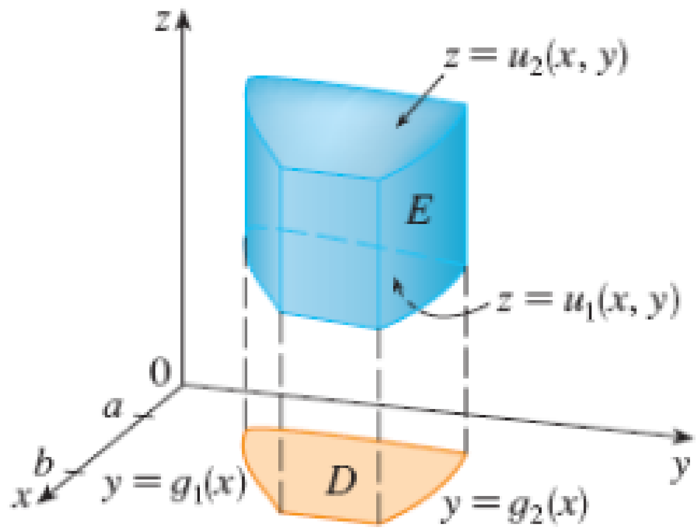


Figure 1.1.2: A type 1 solid region $E$ over a type I plane region $D$

$$\int\int_E\int f(x,y,z)dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)dzdydx.$$

If $D$ is a type I plane region, then the order of integration in the right hand side of the above equation will be as shown (that is, $dzdydx$).
Similarly, if $D$ is a type II plane region, then

$$E = \{(x,y,z)|c \leq y \leq d, h_1(y) \leq x \leq h_2(y), u_1(x,y) \leq z \leq u_2(x,y)\}.$$

Figure 1.1.3 gives an illustration of such a region.
In this case, we will have

$$\int\int_E\int f(x,y,z)dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)dzdxdy.$$

Here, the order of integration in the right hand side of the above equation will be as shown (that is, $dzdxdy$).

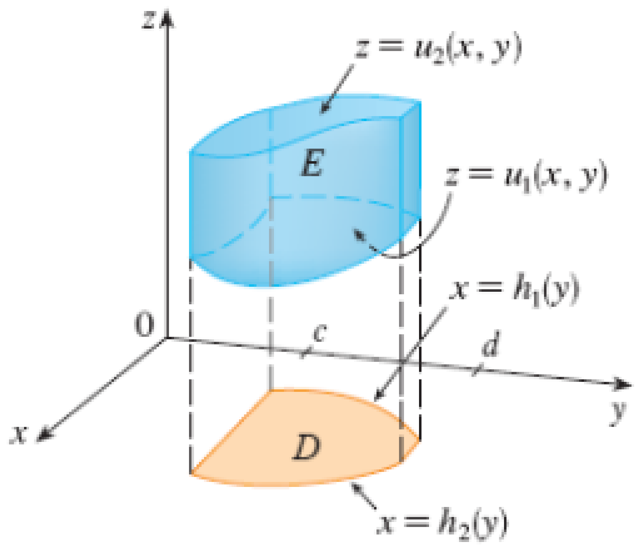


Figure 1.1.3: A type 1 solid region $E$ over a type II plane region $D$

**Remark 1.1.2.** The region $D$ (which we are considering on the $xy$-plane) is nothing but the projection/shadow of the solid $E$ on the $xy$-plane.

**Example 1.1.3.** Evaluate $\int\int_E\int zdV$, where $E$ is the solid tetrahedron bounded by the 4 planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$. The left hand side picture in Figure 1.1.4 depicts the solid region $E$, and the right hand side picture in the same figure depicts the plane region $D$ on the $xy$-plane, which is a projection/shadow of $E$ onto the $xy$-plane.

Here $D = \{(x, y)|0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$, if $D$ is considered as a type I plane region (Although, in this example, $D$ can be considered both as type I and II regions). Also, observe that $0 \leq z \leq 1 - x - y$. So we have

$$E = \{(x, y, z)|0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 1 - x - y\}.$$

Hence

$$\int\int_E\int zdV = \int_{x=0}^{1}\int_{y=0}^{1-x}\int_{z=0}^{1-x-y} zdzdydx.$$

The rest of the calculation is left to the reader as an exercise. □


Figure 1.1.4: Figure for Example 1.1.3

A **type 2** solid region in $\mathbb{R}^3$ is a region $E$ of the form

$$E = \{(x, y, z)|(y, z) \in D, u_1(y, z) \leq x \leq u_2(y, z)\},$$

where $D$ is the projection of $E$ onto the $yz$-plane. Observe that the solid region $E$ (of type 2) is bounded by two surfaces that are graphs of functions of the variables $y$ and $z$. Figure 1.1.5 depicts a type 2 solid region. In this case, we will have

$$\int\int_E\int f(x, y, z)dV = \int_D\int \left[\int_{u_1(y,z)}^{u_2(y,z)} f(x, y, z)dx\right] dA.$$

Similarly, a **type 3** solid region is of the form

$$E = \{(x, y, z)|(x, z) \in D, u_1(x, z) \leq y \leq u_2(x, z)\},$$

where $D$ is the projection of $E$ onto the $xz$-plane. Observe that the solid region $E$ (of type 3) is bounded by two surfaces that are graphs of functions of the variables $x$ and $z$. Figure 1.1.6 depicts a type 3 solid region. In this case, we will have

$$\int\int_E\int f(x, y, z)dV = \int_D\int \left[\int_{u_1(x,z)}^{u_2(x,z)} f(x, y, z)dy\right] dA.$$

**Example 1.1.4.** Evaluate $\int\int_E\int \sqrt{x^2 + z^2}dV$, where $E$ is the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$. Figure 1.1.7 shows the

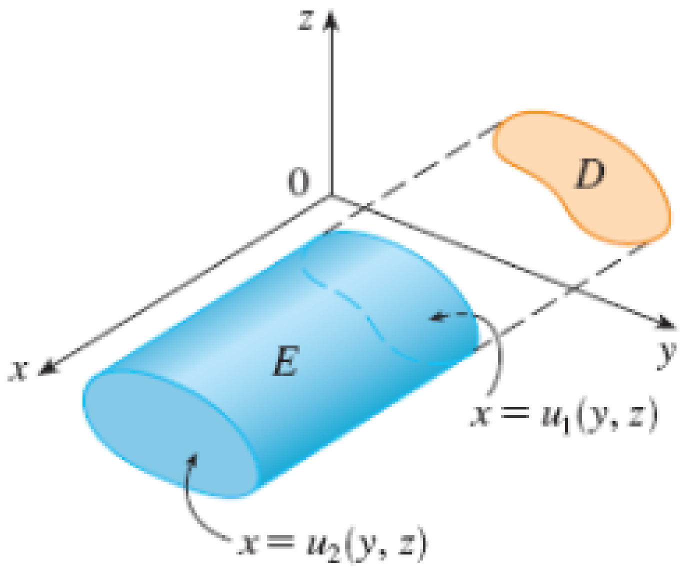


Figure 1.1.5: A type 2 solid region

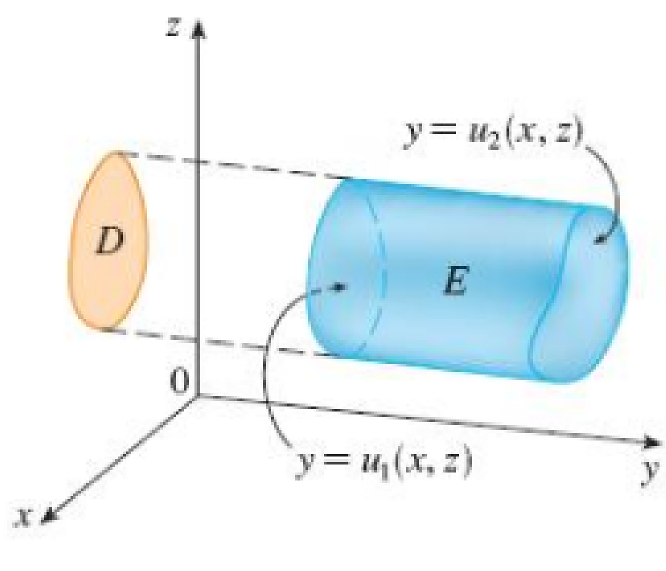


Figure 1.1.6: A type 3 solid region

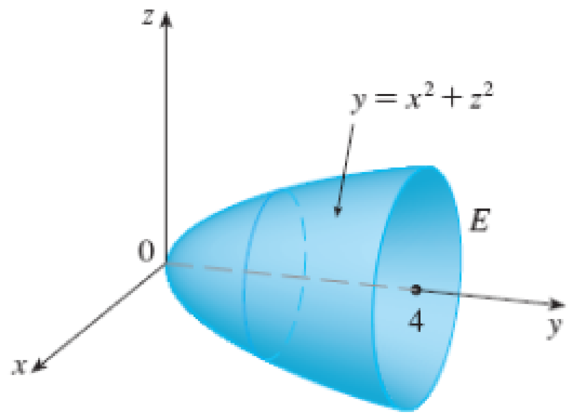


Figure 1.1.7: The region $E$ of Example 1.1.4

solid region $E$.

If we consider $E$ as a type 1 region, then the projection $D_1$ of $E$ will be on the $xy$-plane. Then $D_1$ will be the region in the $xy$-plane as shown in Figure 1.1.8.

Then

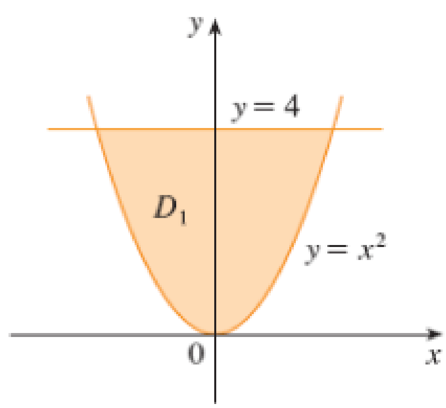


Figure 1.1.8: The region $D_1$ of Example 1.1.4

$$E = \{(x, y, z) | (x, y) \in D_1, -\sqrt{y - x^2} \leq z \leq \sqrt{y - x^2}\},$$

where

$$D_1 = \{(x, y)| - 2 \leq x \leq 2, x^2 \leq y \leq 4\}$$

and then the integration

$$\int_{x=-2}^{2} \int_{y=x^2}^{4} \int_{z=-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2 + z^2} dz dy dx$$

will be very very cumbersome. Similar thing happens if we consider $E$ as a type 2 region.
The easiest way is to consider $E$ as a type 3 region. That is,

$$E = \{(x, y, z)|(x, z) \in D_3, x^2 + z^2 \leq y \leq 4\},$$

where

$$D_3 = \{(x, z)|0 \leq (x, z) \leq 4\}.$$

Figure 1.1.9 shows the region $D_3$. Then

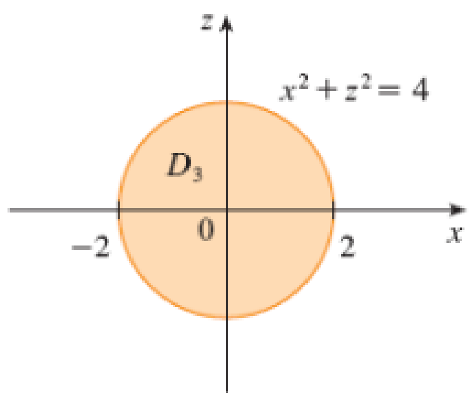


Figure 1.1.9: The region $D_3$ of Example 1.1.4

$$\int\int_E\int \sqrt{x^2 + z^2} dV = \int_{D_3}\int \left[ \int_{y=x^2+z^2}^{4} \sqrt{x^2 + z^2} dy \right] dA$$

$$= \int_{D_3}\int \sqrt{x^2 + z^2}(4 - x^2 - z^2) dA.$$

Our job is going to be easier if we consider $D_3$ as a region in polar coordinates. Observe that, in that case,

$$D_3 = \{(r, \theta) | 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}.$$

Then

$$\int_{D_3}\int \sqrt{x^2 + z^2}(4 - x^2 - z^2) dA = \int_{\theta=0}^{2\pi}\int_{r=0}^{2} r(4 - r^2) r dr d\theta.$$

The rest of the calculation is left to the reader as an exercise. □

**Remark 1.1.5.** In any given problem, according to our convenience, we are going to consider $E$ as a type 1,2 or 3 region or their unions.

**Homework:** Please work out Example 4 on slide number 12 of Lecture 13.

# 2 Triple integrals in polar coordinates

## 2.1 Cylindrical polar coordinates

When we consider the cartesian coordinate system, we simply consider 3 axes, which are orthogonal to each other meeting at the origin $O$. Let us consider a point in space (say, $P(x, y, z)$). If we assume that this point $P$ lies on the surface of a cylinder, then we drop a perpendicular from $P$ onto the $xy$-plane and we join the foot $L$ of the perpendicular with the origin $O$. Figure 2.1.1 shows a cylinder.

Suppose that the line $OL$ makes an angle $\theta$ with the $x$-axis. We are going to assume that the cylinder has radius $r$. If we consider the $xy$-plane and consider the polar coordinate system there, then we can simply write

$$x = r\cos\theta,\ y = r\sin\theta,\ z = z.$$

So we are now changing from the $(x, y, z)$-system to the $(r, \theta, z)$-system and that is nothing but the **cylindrical polar coordinate system**. Please see Figure 2.1.2 for an illustration.

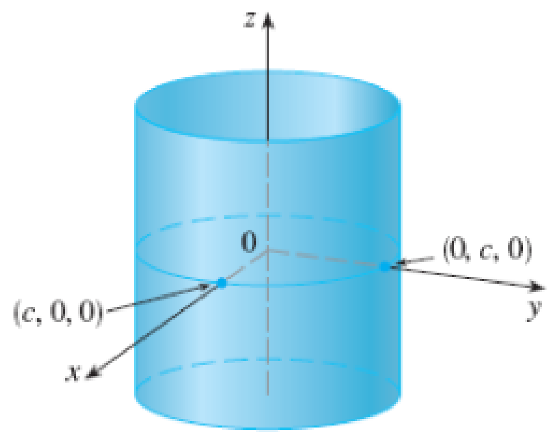


Figure 2.1.1: A cylinder

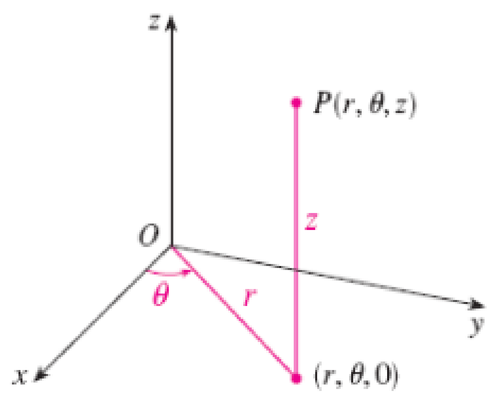


Figure 2.1.2: Cylindrical polar coordinates

## 2.2 Triple integrals in cylindrical polar coordinates

Suppose we want to perform integration in cylindrical polar coordinates, then we are going to consider a polar cylindrical box as shown in blue color in Figure 2.2.1. Let us denote this blue colored box by $E$. Let $D$ denote the shadow or projection of this polar cylindrical box $E$ on the $xy$-plane. If we consider $E$ as a type 1 solid region, then we have

$$E = \{(x, y, z)|(x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}.$$

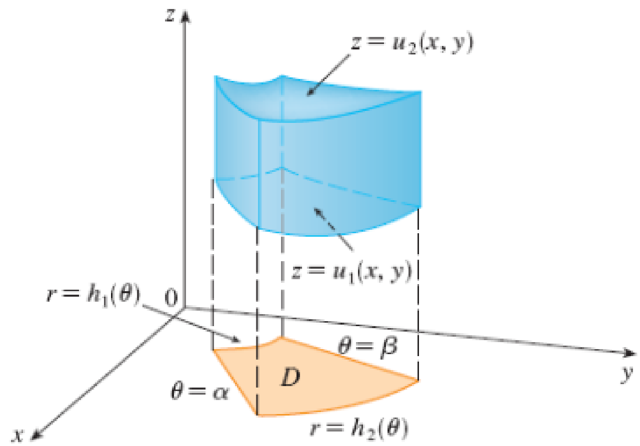


Figure 2.2.1: A polar cylindrical box

If we express the region $D$ in the $xy$-plane as a polar rectangle, then we have

$$D = \{(r, \theta) | \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}.$$

Now,

$$\int \int_E \int f(x, y, z) dV = \int_D \int \left[ \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) dz \right] dA.$$

If we perform the integration of the expression within paranthesis in the above equation, that is, if we perform the integration $\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) dz$, then we are going to get a function of $x$ and $y$. This function of $x$ and $y$ can be transformed again into a function of $r$ and $\theta$ and eventually it will lead us to

$$\int \int_E \int f(x, y, z) dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta, r\sin\theta)}^{u_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z) r dz dr d\theta.$$

Here $0 \leq \beta - \alpha \leq 2\pi$.

**Remark 2.2.1.** We are not going to be bothered about how did we obtain $dV = r dz dr d\theta$ because later on, we are going to study a general methodology about transformation of coordinates, where we will see how to tackle any kind of transformation.

**Remark 2.2.2.** If $f(x, y, z) = 1$, then $\int \int_E \int f(x, y, z) dV = \int \int_E \int dV$ and this integral simply represents the volume of $E$.

**Example 2.2.3.** A solid $E$ lies within the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. The density at any point is proportional to its distance from the axis of the cylinder. Find the mass of $E$. Figure 2.2.2 shows the solid $E$.
We know that

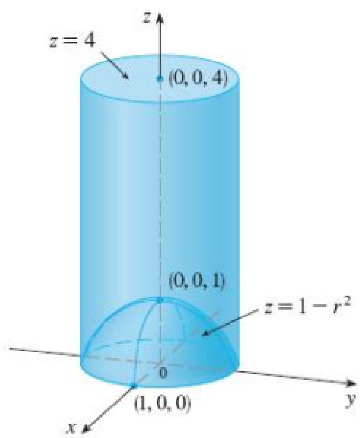


Figure 2.2.2: Figure for Example 2.2.3

$$\text{Mass=Density} \times \text{Volume.}$$

So to find the mass of $E$, we will have to integrate the density function over the entire region $E$. It is given that the density $\rho$ at any point $P(x, y, z)$ is proportional to its distance $\sqrt{x^2 + y^2}$ from the $z$-axis. Then we have

$$\rho = k\sqrt{x^2 + y^2},$$

for some constant $k$. Then

$$\text{Mass} = \int\int_E\int \rho dV = \int\int_E\int k\sqrt{x^2 + y^2}dV$$

$$= k\int_D\int \left[\int_{z=1-x^2-y^2}^{4} \sqrt{x^2 + y^2}dz\right] dA,$$

where $D$ is the circular disk given by

$$D = \{(r, \theta)|0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}.$$

Hence

$$\text{Mass} = k\int_D\int\left[\int_{z=1-x^2-y^2}^{4}\sqrt{x^2+y^2}dz\right]dA = k\int_D\int\sqrt{x^2+y^2}[3+x^2+y^2]dA$$

$$= k\int_{\theta=0}^{2\pi}\int_{r=0}^{1} r(3+r^2)rdrd\theta.$$

The rest of the calculation is left to the reader as an exercise. □

# 3 Triple integrals in spherical coordinates

Suppose we consider the cartesian coordinate system and a point $P(x, y, z)$ in space. Imagine that this point $P(x, y, z)$ lies on a sphere of radius $\rho$.

Let us draw a perpendicular from $P$ onto the $xy$-plane. Let $P'$ denote the

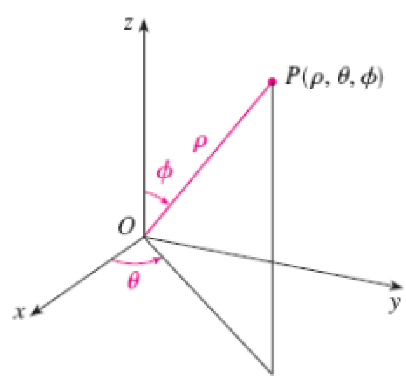


Figure 3.0.1: Spherical coordinates

foot of the perpendicular. The coordinates of $P'$ are $(x, y, 0)$. Let us join the origin $O$ with $P'$ and also assume that the radius vector $\overrightarrow{OP}$ makes an angle $\phi$ with the $z$-axis. Observe that the radius vector $\overrightarrow{OP}$ can traverse an angle (with the $z$-axis) whose maximum extent is $\pi$. So the angle $\phi$ can take values in the interval $[0, \pi]$. Let $\overrightarrow{OP'}$ make an angle $\theta$ with the $x$-axis. Clearly then the extent of the angle $\theta$ is from 0 to $2\pi$. Then we have

$$x = r\cos\theta,\ y = r\sin\theta,$$

where $\overrightarrow{OP'} = r$. Please see Figure 3.0.2 for an illustration. If we consider

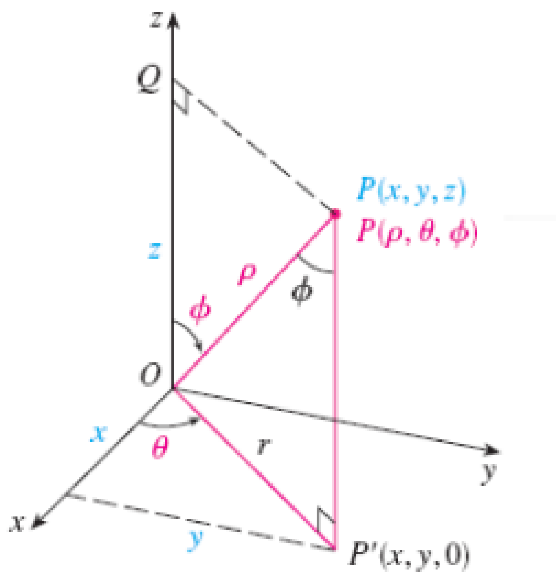


Figure 3.0.2: Relation between Spherical and Cartesian coordinates

the triangle $OPP'$, then we can easily see that it is a right angled triangle in which the angle $POP'$ equals $\frac{\pi}{2} - \phi$, $|\overrightarrow{OP'}| = r$, and $|\overrightarrow{OP}| = \rho$. From triangle $OPP'$, we get

$$\frac{r}{\rho} = \cos(\frac{\pi}{2} - \phi)$$
$$\Rightarrow r = \rho \sin\phi.$$

Then we have

$$x = r\cos\theta = \rho\sin\phi\cos\theta$$

and

$$y = r\sin\theta = \rho\sin\phi\sin\theta.$$

Also, $|\overrightarrow{PP'}| = z$ and $\frac{z}{\rho} = \sin(\frac{\pi}{2} - \phi)$. This implies that $z = \rho\cos\phi$. In this way, we can convert the cartesian coordinate system into spherical coordinate system.

In the spherical coordinate system, the counterpart of a rectangular box is a **spherical wedge** given by

$$E = \{(\rho, \theta, \phi) | a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\},$$

where $a \geq 0$, $\beta - \alpha \leq 2\pi$ and $d - c \leq \pi$. A spherical wedge is shown in blue color in Figure 3.0.3.

Let us divide the entire sphere into elementary spherical wedges. Let $\Delta V_{ijk}$

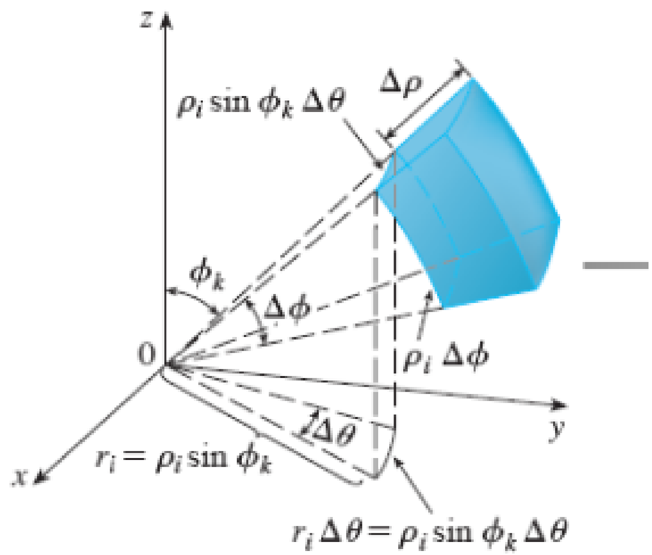


Figure 3.0.3: A spherical wedge

be the volume of one such elementary spherical wedge. Let $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ be a sample point in this elementary spherical wedge, whose volume is $\Delta V_{ijk}$. Then we have

$$\iiint_E f(x, y, z)\, dV = \lim_{l,m,n \to \infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\, \Delta V_{ijk}$$

$$= \lim_{l,m,n \to \infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(\tilde{\rho}_i \sin\tilde{\phi}_k \cos\tilde{\theta}_j,\ \tilde{\rho}_i \sin\tilde{\phi}_k \sin\tilde{\theta}_j,\ \tilde{\rho}_i \cos\tilde{\phi}_k)\ \tilde{\rho}_i^2 \sin\tilde{\phi}_k\, \Delta\rho\, \Delta\theta\, \Delta\phi$$

But this sum is a Riemann sum for the function

$$F(\rho, \theta, \phi) = f(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)\, \rho^2 \sin\phi$$

Consequently, we have arrived at the following **formula for triple integration in spherical coordinates**.

$\boxed{3}$ $$\iiint_E f(x, y, z)\, dV$$

$$= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)\, \rho^2 \sin\phi \; d\rho\, d\theta\, d\phi$$

where $E$ is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid a \leqslant \rho \leqslant b,\ \alpha \leqslant \theta \leqslant \beta,\ c \leqslant \phi \leqslant d\}$$

The volume element $dV$ in spherical coordinates is given by $dV = \rho^2 \sin\phi d\rho d\theta d\phi$. Figure 3.0.4 provides an illustration of the volume element $dV$.

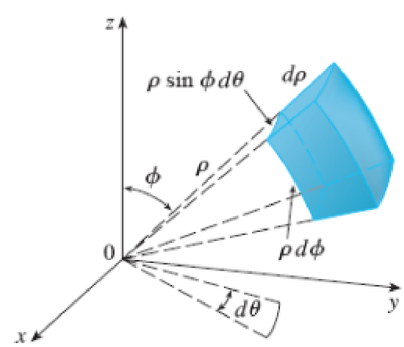


Figure 3.0.4: The volume element $dV$

**Homework:** Please work out example 4 on slide number 5 of Lecture 14.

————–-END————–